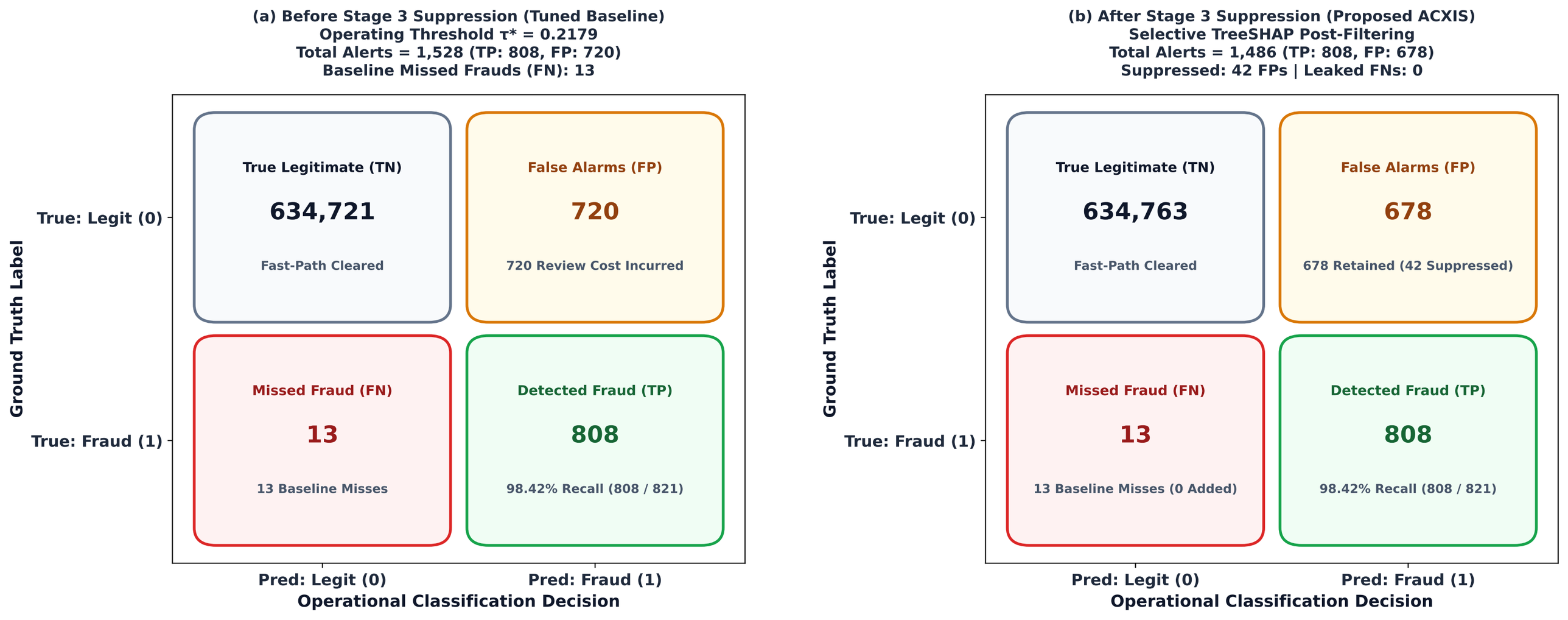
(a) Before Stage 3 Suppression (Tuned Baseline)
Operating Threshold τ* = 0.2179
Total Alerts = 1,528 (TP: 808, FP: 720)
Baseline Missed Frauds (FN): 13
True Legitimate (TN)
634,721
Fast-Path Cleared
False Alarms (FP)
720
720 Review Cost Incurred
Missed Fraud (FN)
13
13 Baseline Misses
Detected Fraud (TP)
808
98.42% Recall (808 / 821)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Classification Decision
(b) After Stage 3 Suppression (Proposed ACXIS)
Selective TreeSHAP Post-Filtering
Total Alerts = 1,486 (TP: 808, FP: 678)
Suppressed: 42 FPs | Leaked FNs: 0
True Legitimate (TN)
634,763
Fast-Path Cleared
False Alarms (FP)
678
678 Retained (42 Suppressed)
Missed Fraud (FN)
13
13 Baseline Misses (0 Added)
Detected Fraud (TP)
808
98.42% Recall (808 / 821)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Classification Decision